

Problem Set 3: Measures of Location, Dispersion, and Skewness

Statistics 1: Data Exploration

Question Set

Q1. Effect of Linear Transformation on Mean and Variance

Let X be a random variable with mean $\mu_X = 20$ and variance $\sigma_X^2 = 25$. Define a new variable $Y = 3X + 5$.

1. Find $E(Y)$ and $\text{Var}(Y)$.
2. Explain in general how mean and variance change under a linear transformation $Y = aX + b$.

Q2. Combined Mean and Variance

Two sections of a class have the following statistics on test scores:

Section	$\bar{x}$	s	n
A	72	8	25
B	68	6	35

1. Compute the combined mean $\bar{x}_c$ and combined variance using

$$\bar{x}_c = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}, \quad s_c^2 = \frac{n_1(s_1^2 + (\bar{x}_1 - \bar{x}_c)^2) + n_2(s_2^2 + (\bar{x}_2 - \bar{x}_c)^2)}{n_1 + n_2}.$$

2. Comment on the overall variability of the combined class.

Q3. Relation Between Mean, Median, and Mode

A moderately skewed distribution has mean $\bar{x} = 62$ and standard deviation $s = 10$. If Pearson's coefficient of skewness is $S_k = 0.6$, find the mode and median using the relations:

$$S_k = \frac{\bar{x} - \text{Mode}}{s}, \quad \bar{x} - \text{Mode} = 3(\bar{x} - \text{Median}).$$

Q4. Moment-Based Measures of Skewness and Kurtosis

The first four central moments of a distribution are:

$$\mu'_1 = 0, \quad \mu_2 = 16, \quad \mu_3 = 32, \quad \mu_4 = 500.$$

1. Compute the coefficients:

$$\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}}, \quad \beta_2 = \frac{\mu_4}{\mu_2^2}.$$

2. Comment on the symmetry and peakedness of the distribution.

Q5. Comparison of Variability Using Coefficient of Variation

Two machines produce metal rods:

Machine	$\bar{x}$	SD
A	50.2	0.6
B	49.8	0.4

1. Compute the coefficient of variation:

$$CV = \frac{SD}{\bar{x}} \times 100.$$

2. Determine which machine is more consistent.
3. If rework cost increases with variability, which machine is preferable?

Q6. Relation Between Mean, Median, and Mode — Interpretation

For a certain dataset, the mean and median are 45 and 50 respectively.

1. Estimate the mode using

$$\text{Mode} = 3(\text{Median}) - 2(\text{Mean}).$$

2. Indicate whether the distribution is positively or negatively skewed.

Q7. Variance from a Frequency Distribution

The following table gives the frequency distribution of the number of books issued per day in a library:

x	2	4	6	8	10
f	3	7	12	8	5

Compute the mean and variance using the step-deviation method.

Comment on the degree of dispersion.

Q8. Effect of Change of Origin and Scale on Standard Deviation

Let the standard deviation of a variable X be $\sigma_X = 5$. Define a new variable $Y = \frac{X-10}{2}$.

1. Find σ_Y .
2. Derive the general relationship between the standard deviations of X and Y when $Y = aX + b$.

Q9. Median and Quartile Deviation (Grouped Data)

The following table gives the wages (in) of workers in a factory:

Class Interval	0–20	20–40	40–60	60–80	80–100
Frequency	5	8	15	16	6

Find the median using

$$\text{Median} = L + \left(\frac{\frac{N}{2} - c.f.}{f} \right) \times h.$$

Compute the first and third quartiles and the quartile deviation.

Interpret the result in the context of wage variability.

Q10. Karl Pearson's and Bowley's Skewness — Comparison

The monthly incomes (in) of 10 families are given below:

12, 15, 20, 22, 25, 30, 32, 35, 40, 45.

1. Compute the mean, median, and mode.
2. Find Karl Pearson's coefficient of skewness:

$$Sk_P = \frac{\bar{x} - \text{Mode}}{SD}.$$

3. Find Bowley's coefficient of skewness:

$$Sk_B = \frac{Q_3 + Q_1 - 2 \times \text{Median}}{Q_3 - Q_1}.$$

4. Compare both measures and comment on the nature of the distribution.

End of Problem Set 3